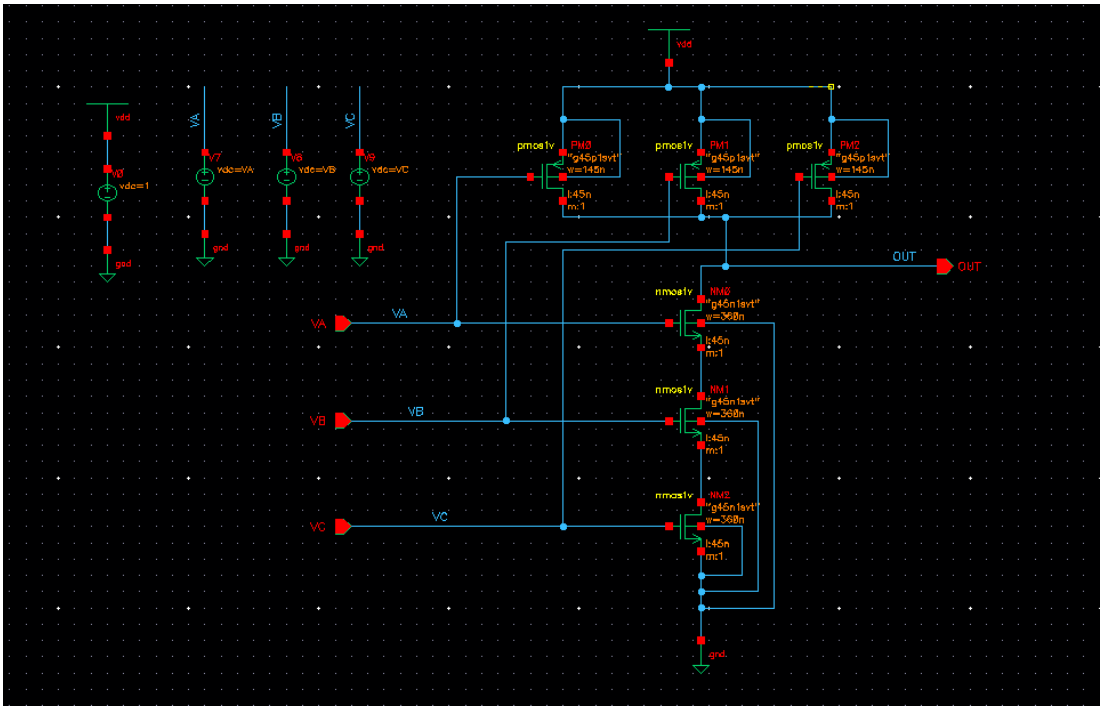
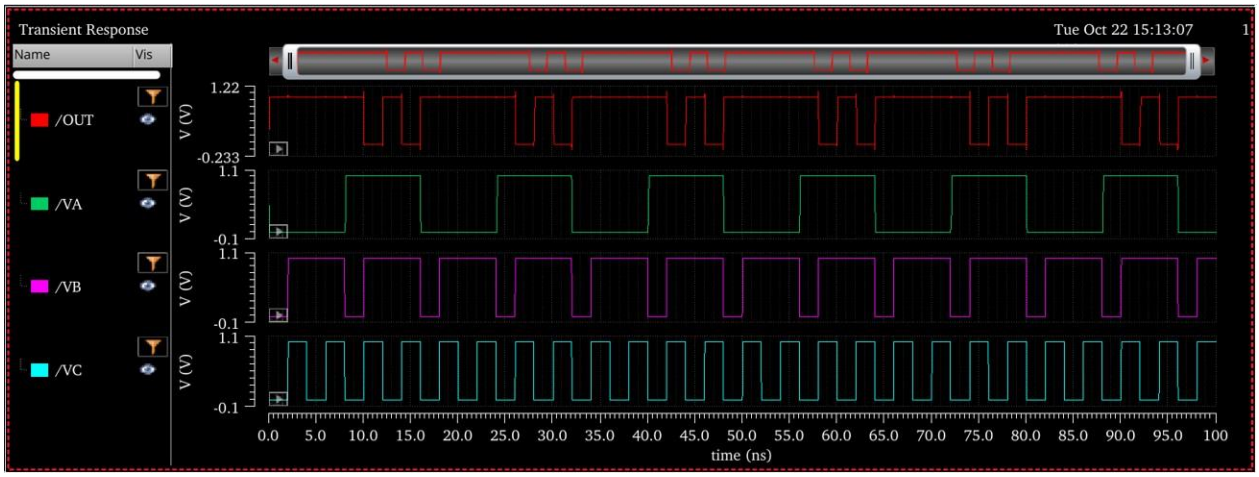
CMOS Logic Design and Domino Logic Analysis Using Cadence Virtuoso

1) Perform the VTC analysis of a three-input NAND and NOR gate and sketch its data-dependent. Size the NMOS devices PMOS devices for unit inverter delay in 45nm Technology.

3-input NAND circuit

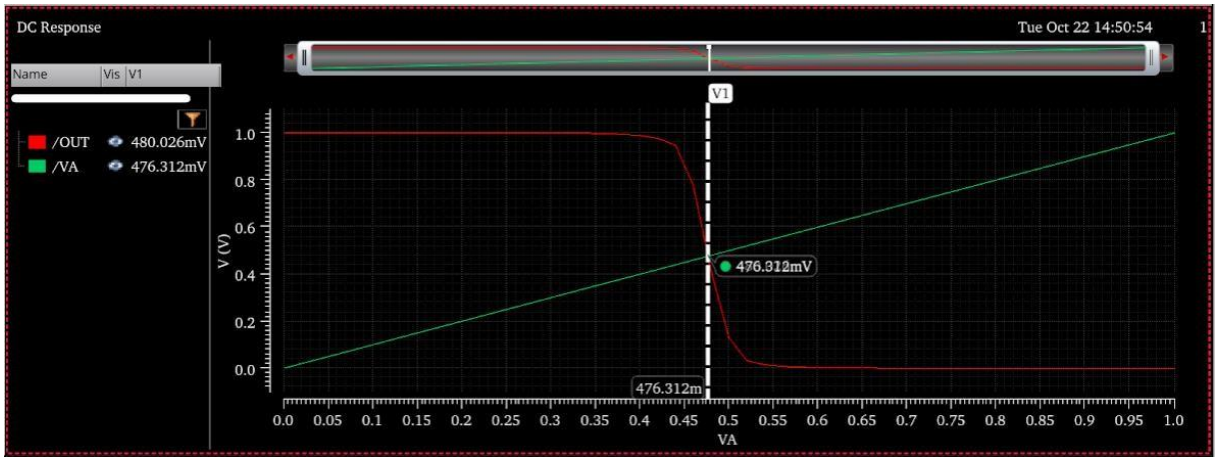


Transient Analysis:

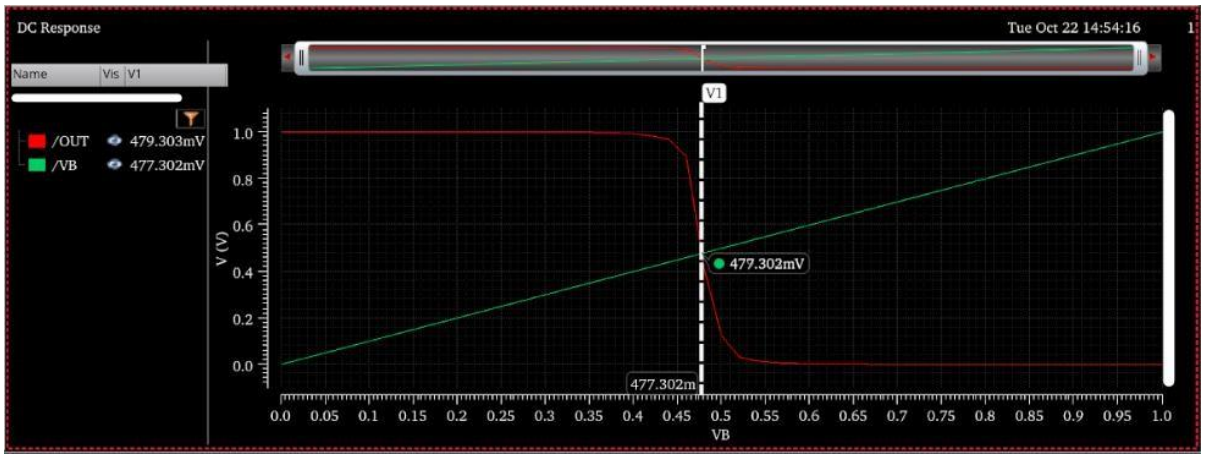


DC Responses:

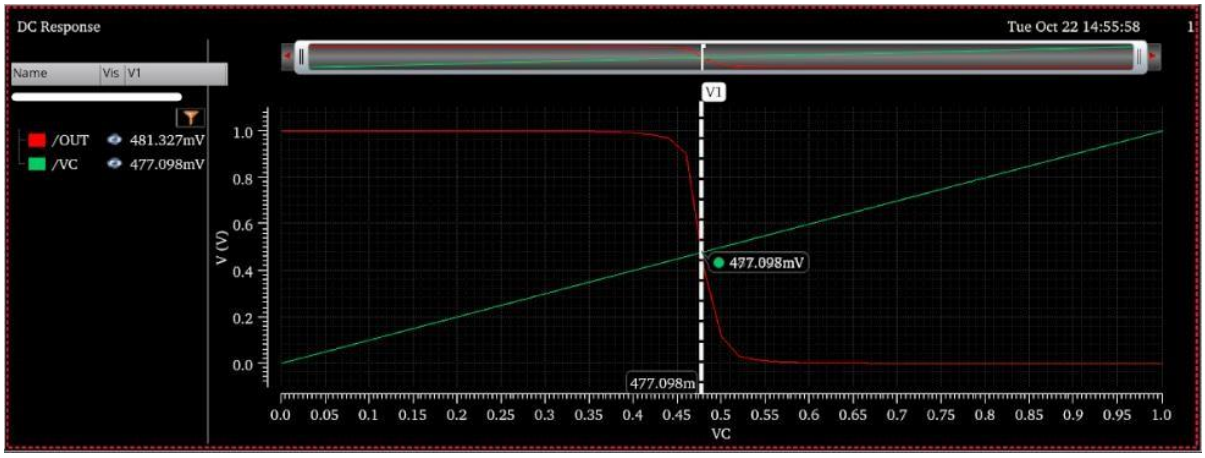
1) VA



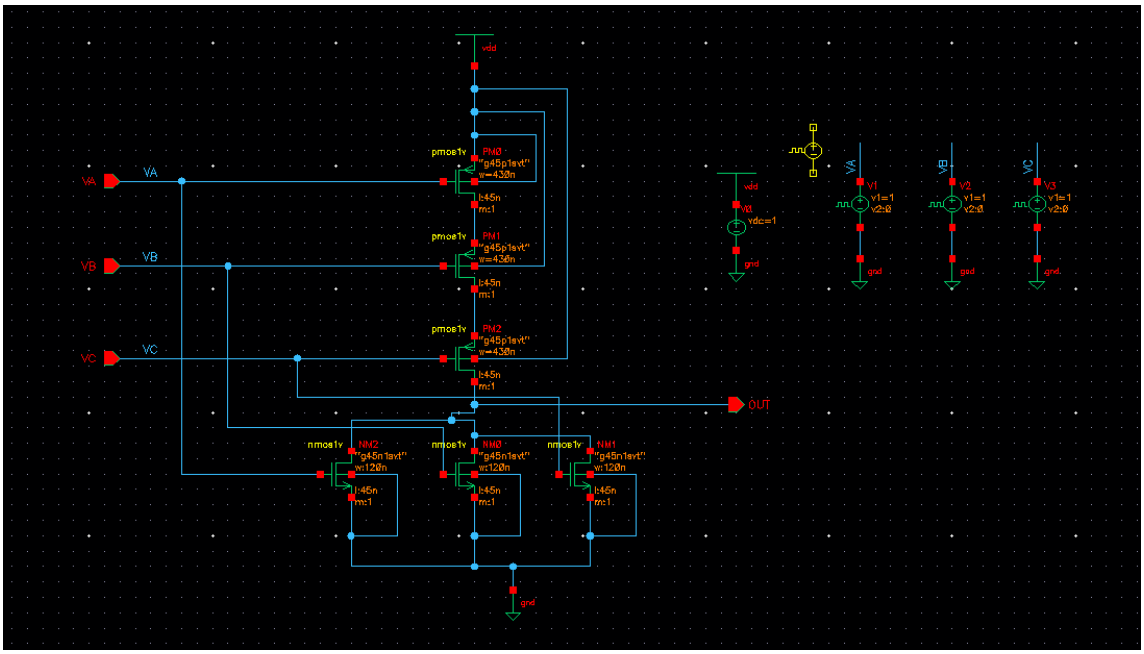
2) VB



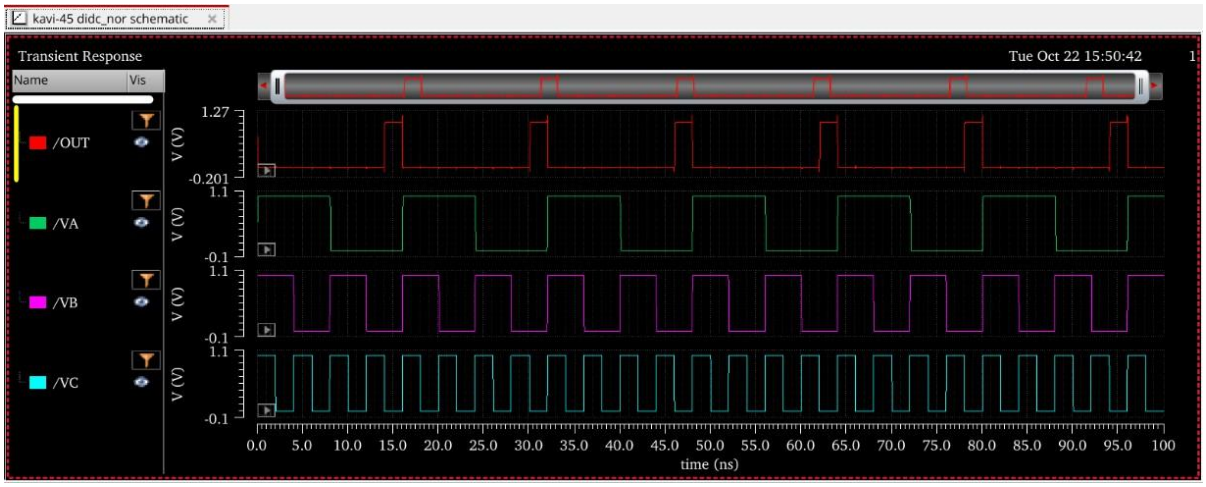
3) VC



3-input NOR circuit

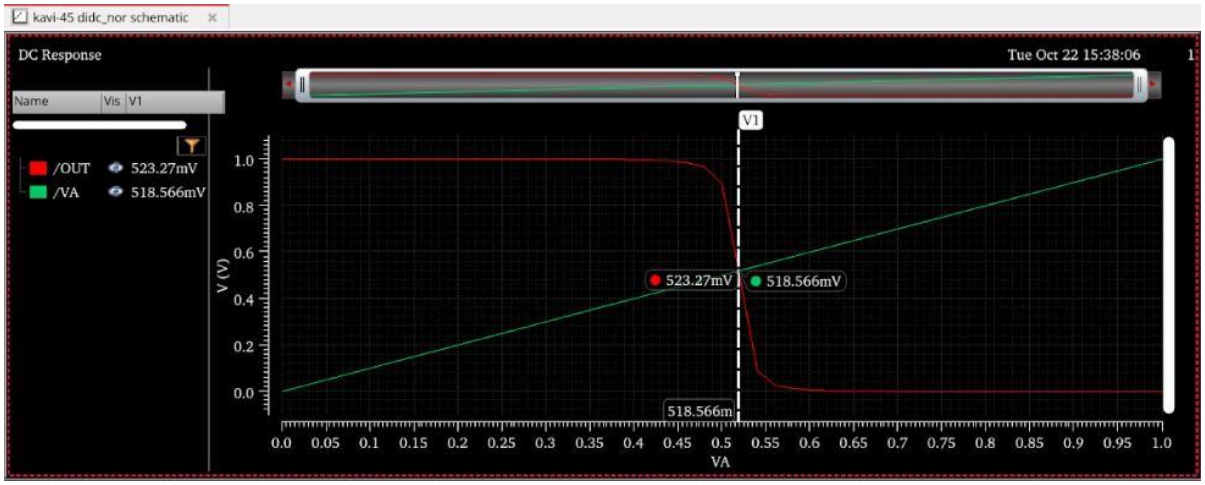


Transient Analysis:

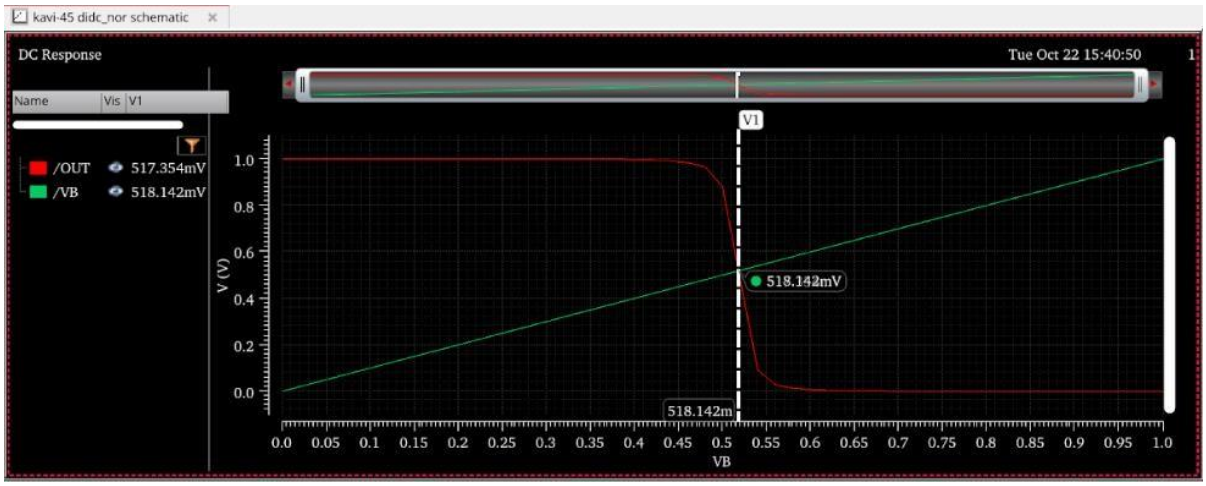


DC Responses:

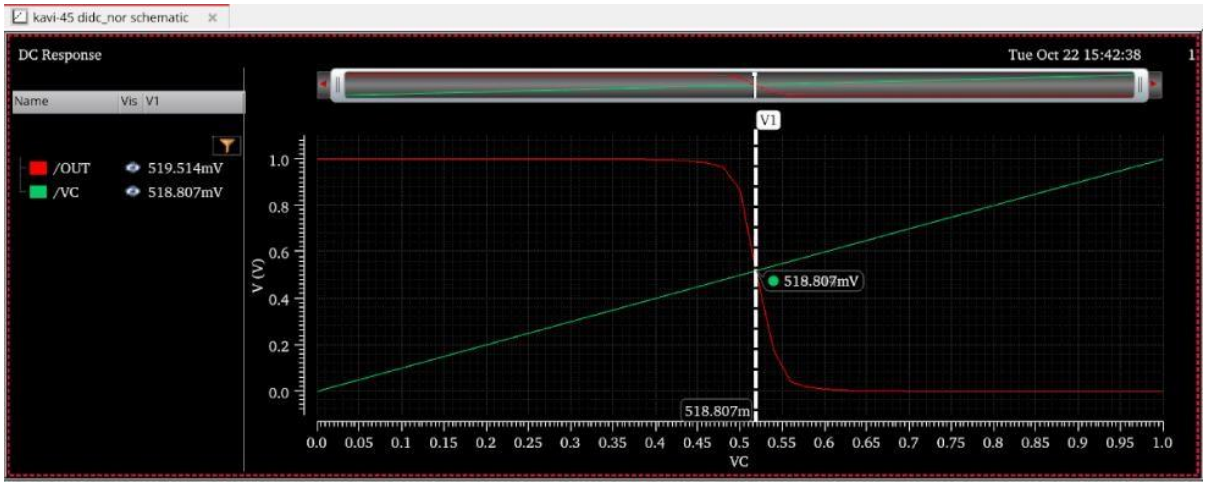
1) VA



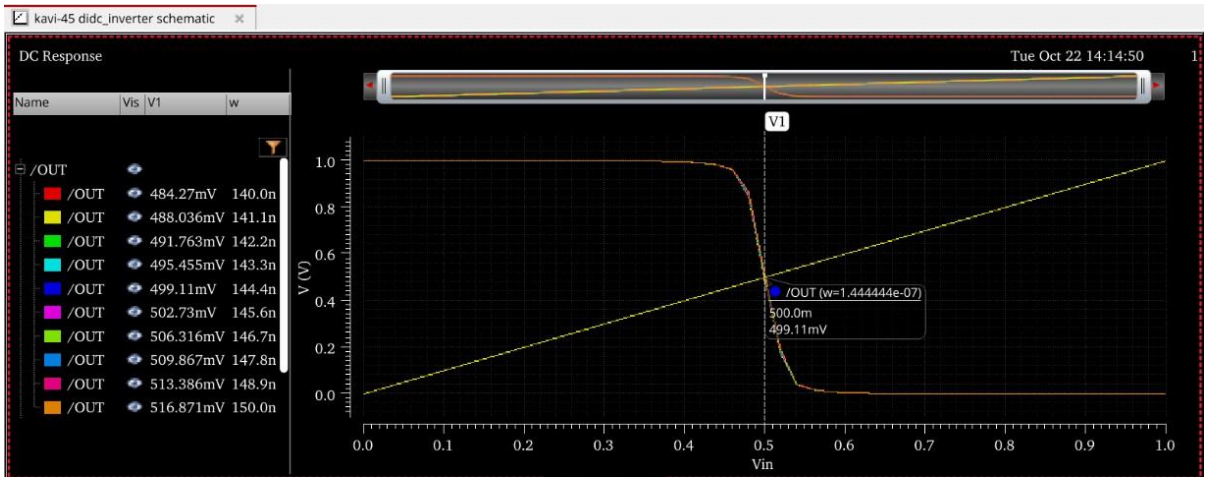
2) VB



3) VC



CMOS Inverter – Parametric Analysis



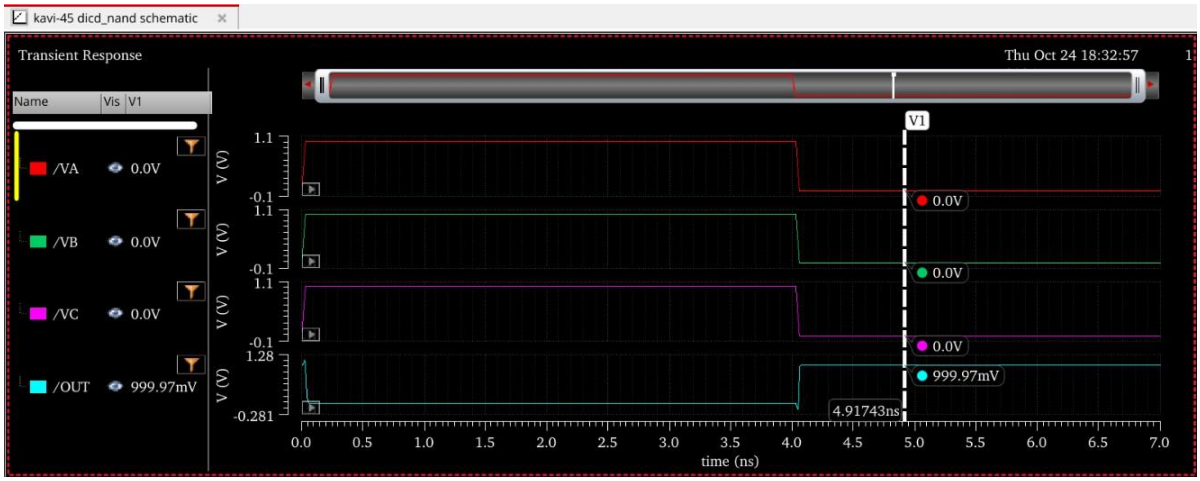
Width (Wp) = 144n

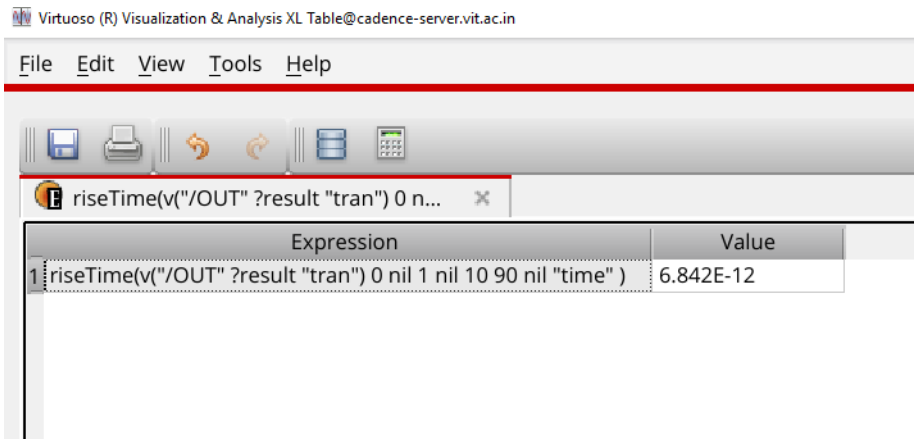
---

2) Analyse the pull up and Pull down delay for the same three-input NAND and NOR gate for different data dependencies of 000 to 111.

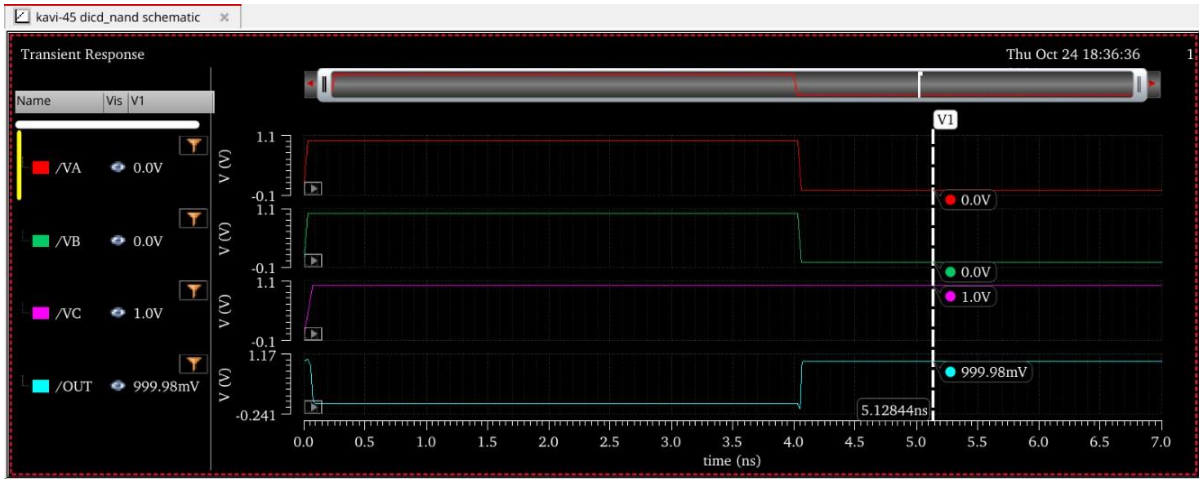
3-input NAND

i) 000





ii) 001



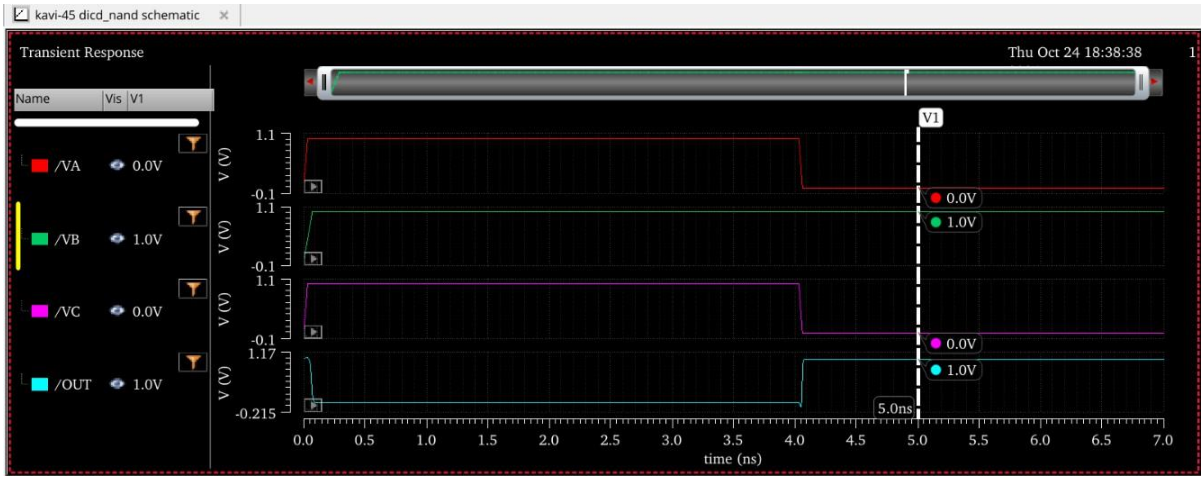
Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in

File Edit View Tools Help

riseTime(v("/OUT" ?result "tran") 0 n...

	Expression	Value
1	riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")	8.416E-12

iii) 010



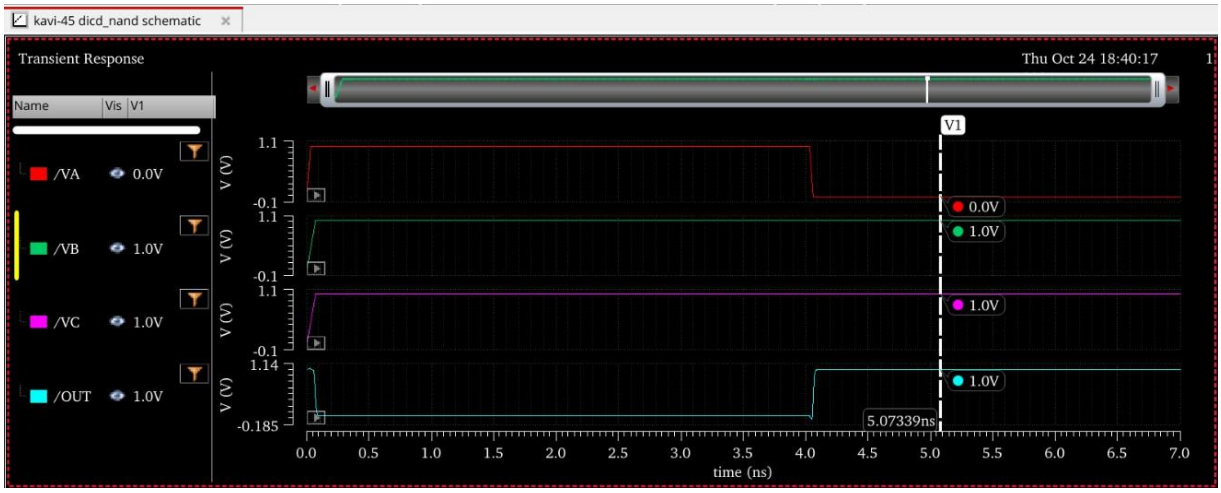
Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in

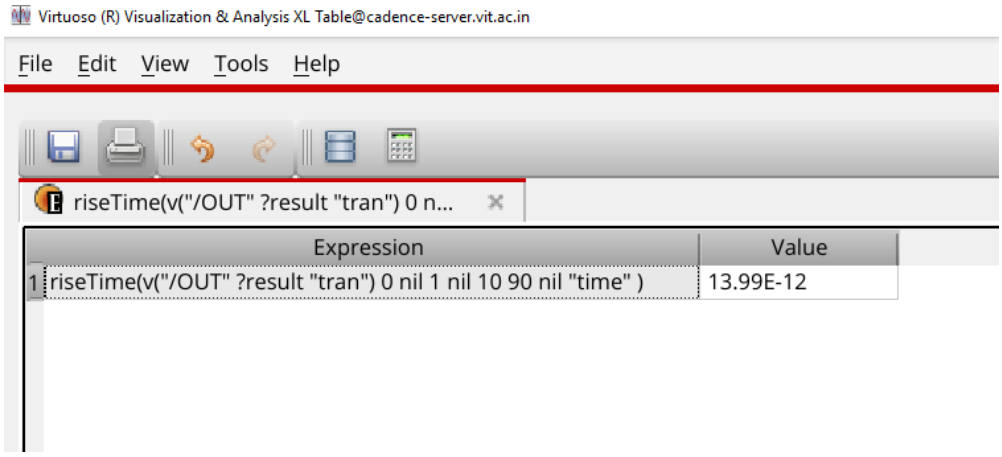
File Edit View Tools Help

riseTime(v("/OUT" ?result "tran") 0 n...

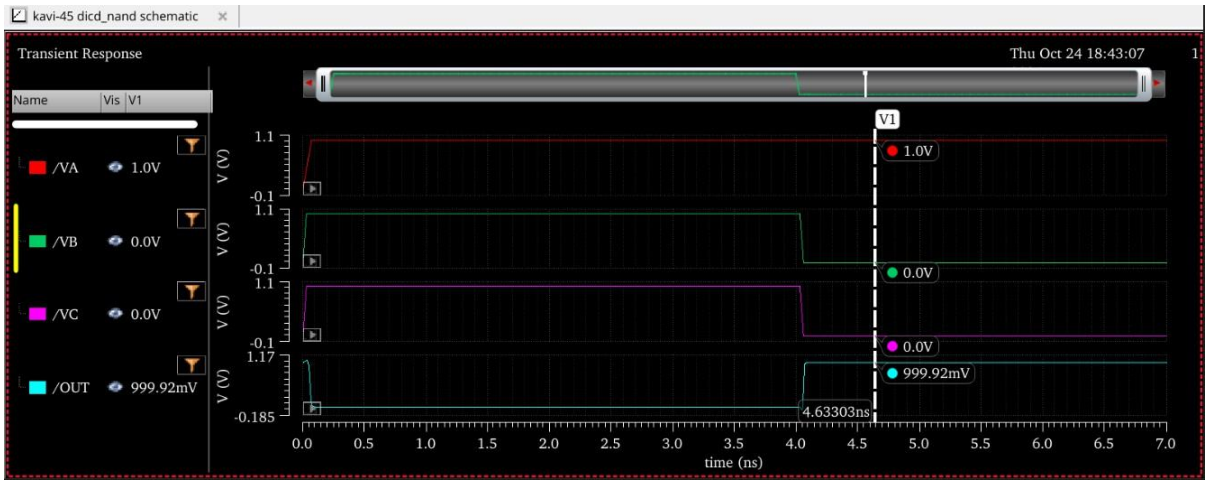
	Expression	Value
1	riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")	8.659E-12

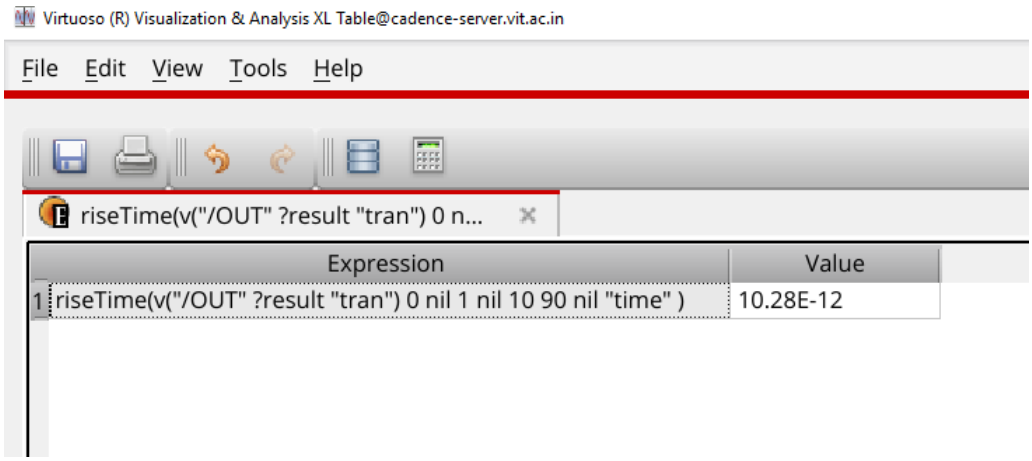
iv) 011


kavi-45 dicd_nand schematic
Transient Response
Thu Oct 24 18:40:17
Name
Vis
V1
/VA
0.0V
/VB
1.0V
/VC
1.0V
/OUT
1.0V
V (V)
1.1
-0.1
1.14
-0.185
V1
0.0V
1.0V
1.0V
1.0V
5.07339ns
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
time (ns)

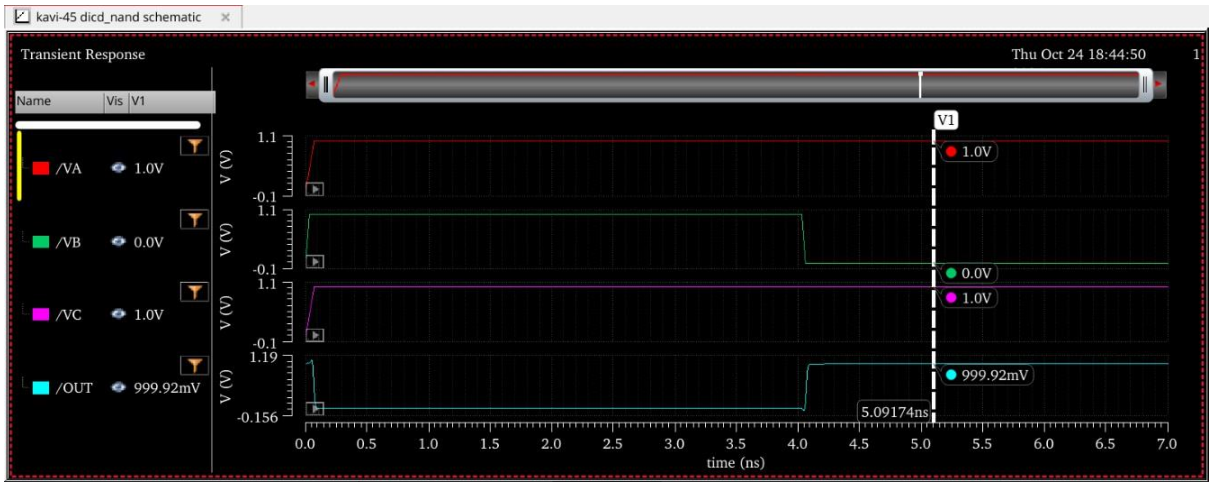

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File
Edit
View
Tools
Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1
riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
13.99E-12

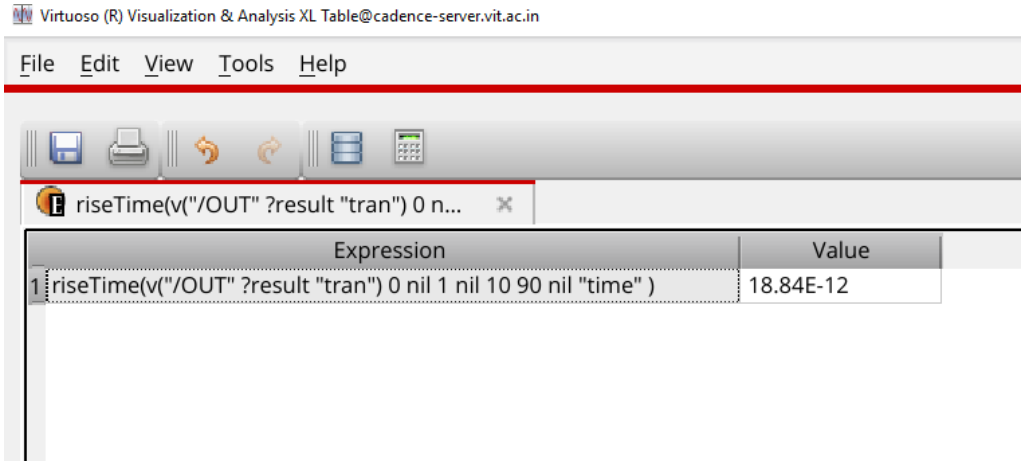
v) 100


kavi-45 dicd_nand schematic
Transient Response
Thu Oct 24 18:43:07
Name
Vis
V1
/VA
1.0V
/VB
0.0V
/VC
0.0V
/OUT
999.92mV
V (V)
1.1
-0.1
1.17
-0.185
V1
1.0V
0.0V
0.0V
999.92mV
4.63303ns
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
time (ns)

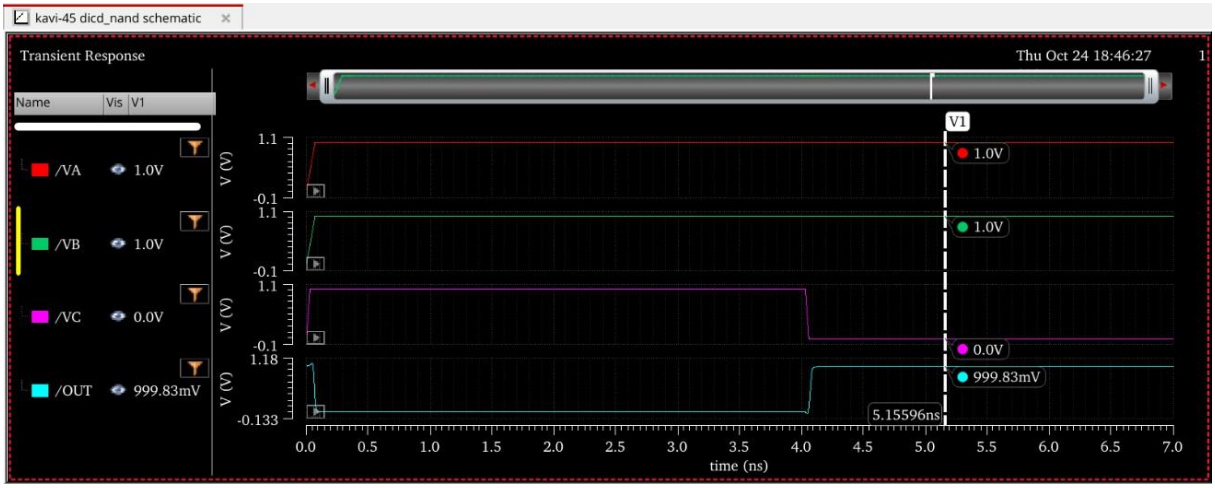

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
10.28E-12

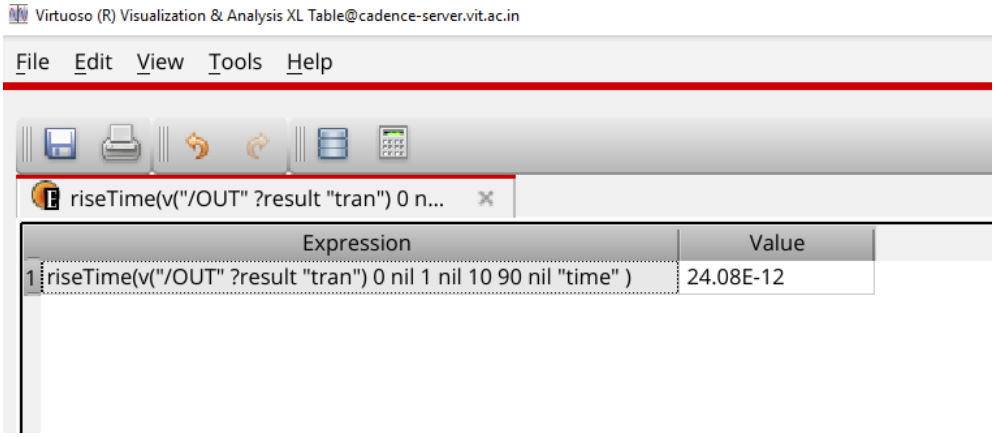
vi) 101


kavi-45 dicd_nand schematic
Transient Response
Thu Oct 24 18:44:50
Name
Vis
V1
/VA
1.0V
/VB
0.0V
/VC
1.0V
/OUT
999.92mV
V (V)
1.1
-0.1
1.19
-0.156
V1
1.0V
0.0V
1.0V
999.92mV
5.09174ns
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
time (ns)

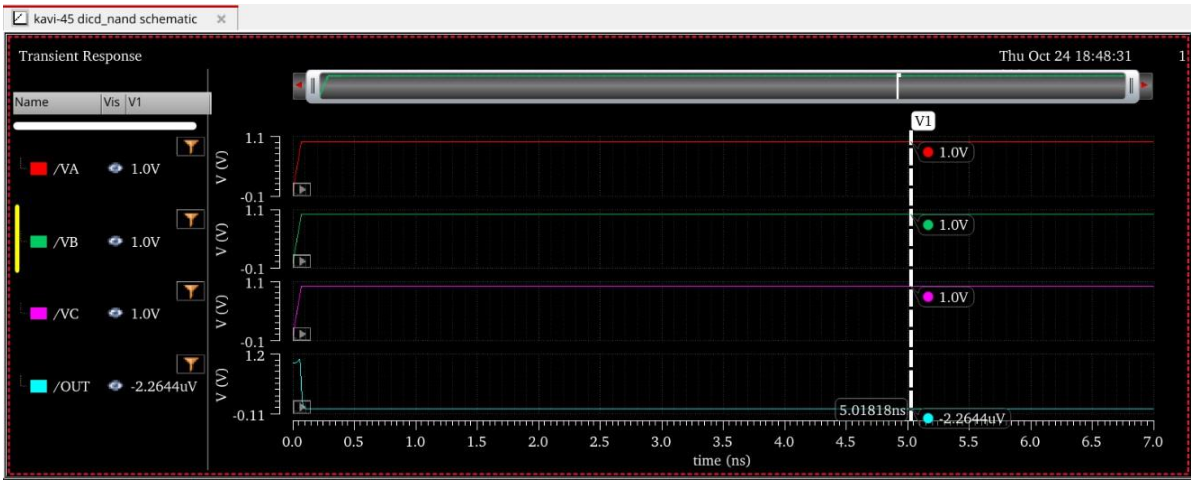

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
18.84E-12

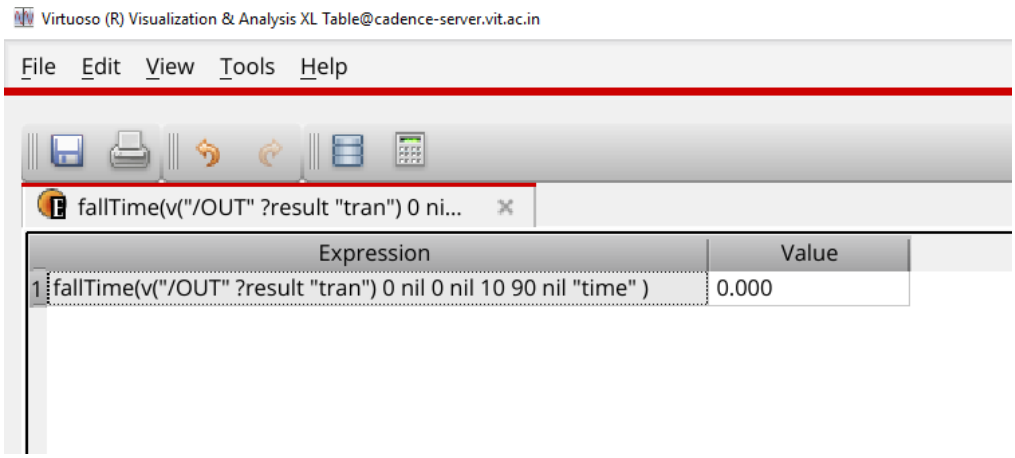
vii) 110





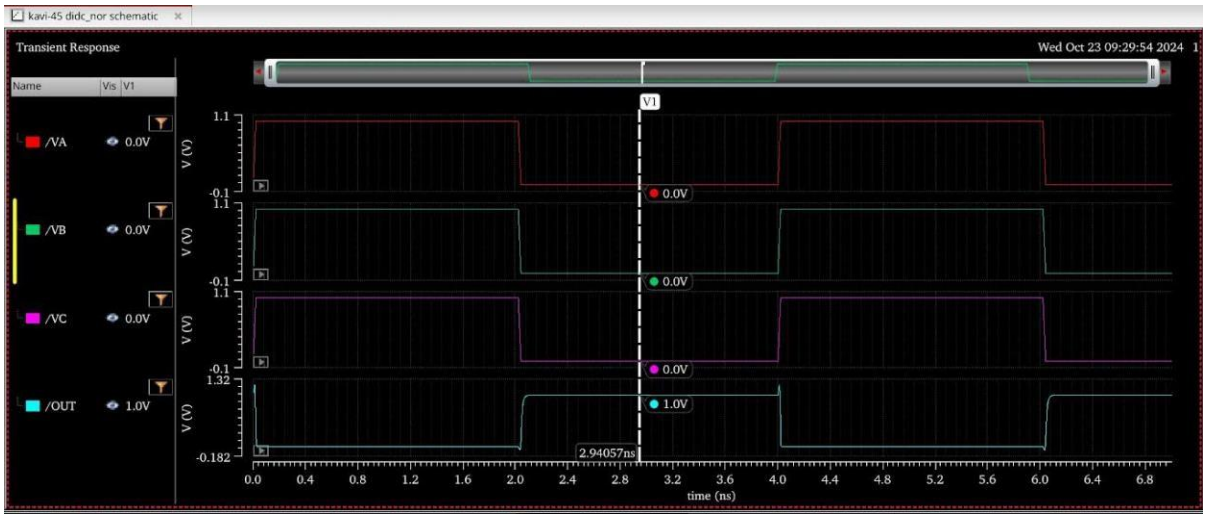
viii) 111

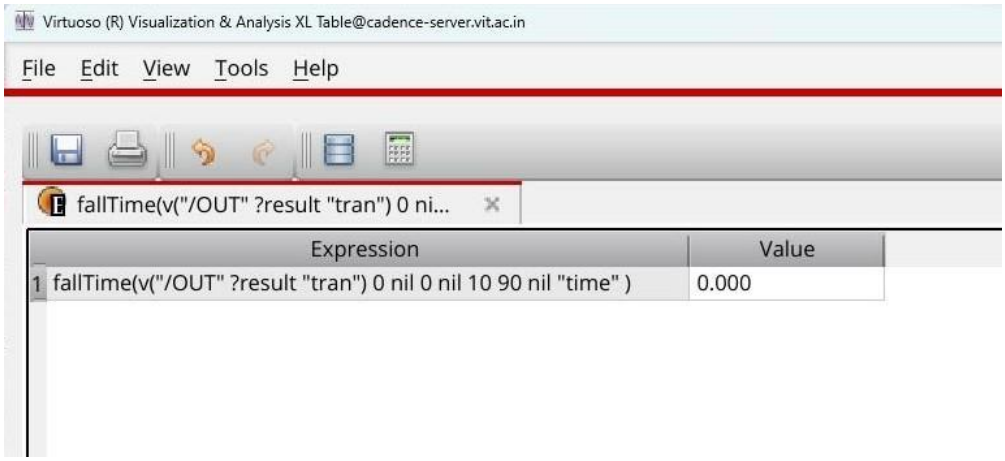




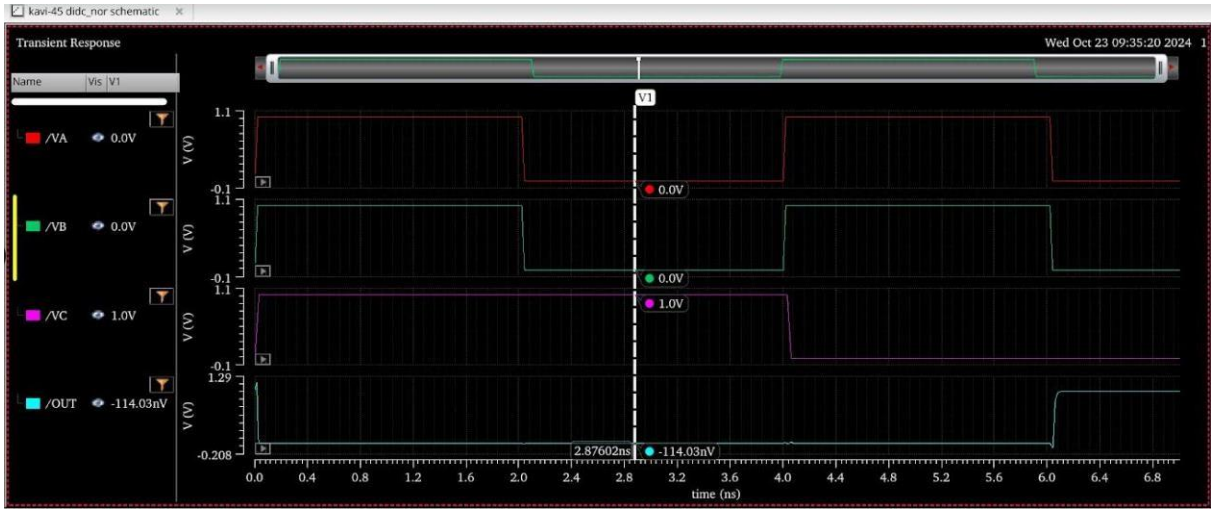
3-input NOR

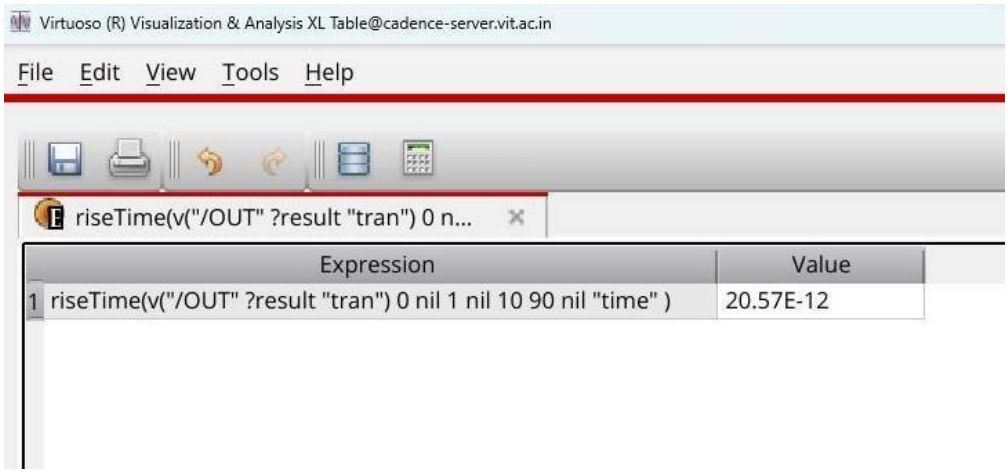
i) 000



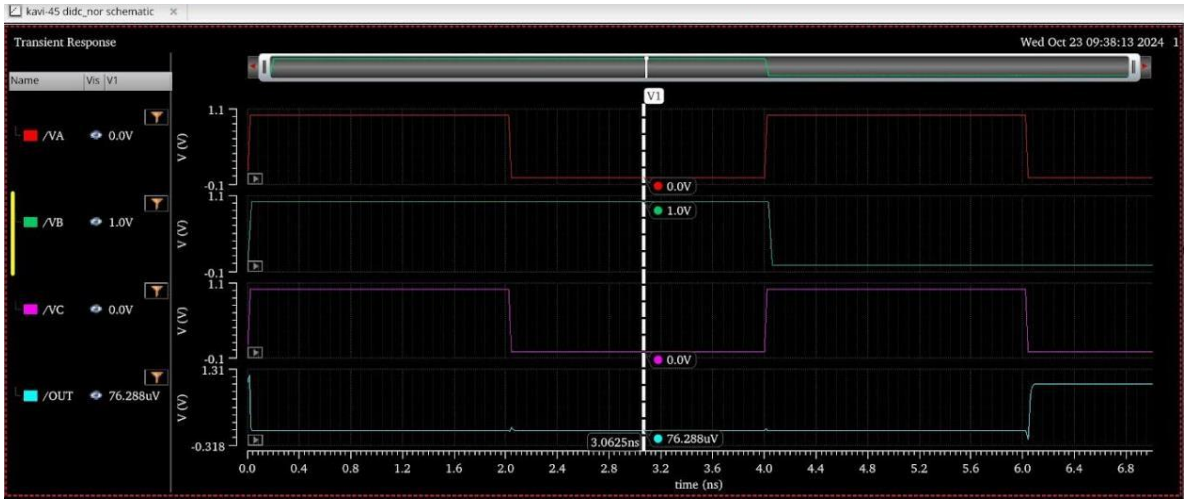


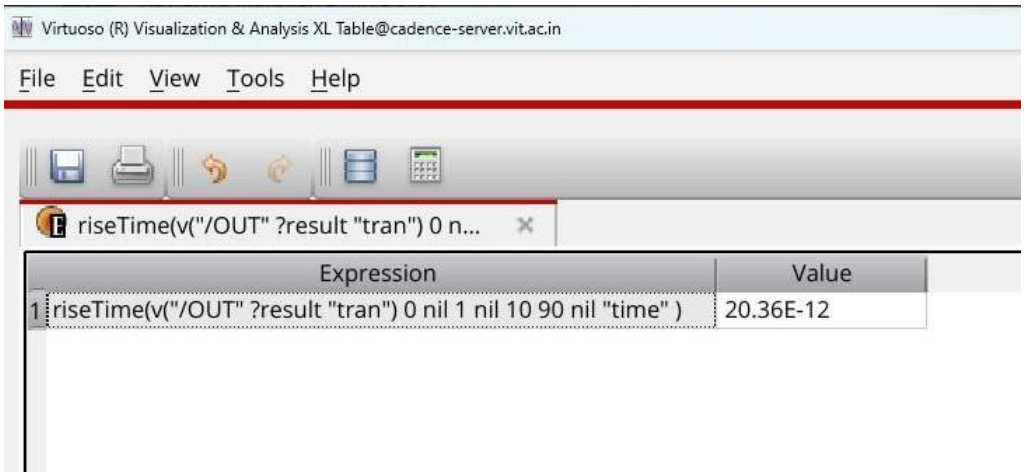
ii) 001



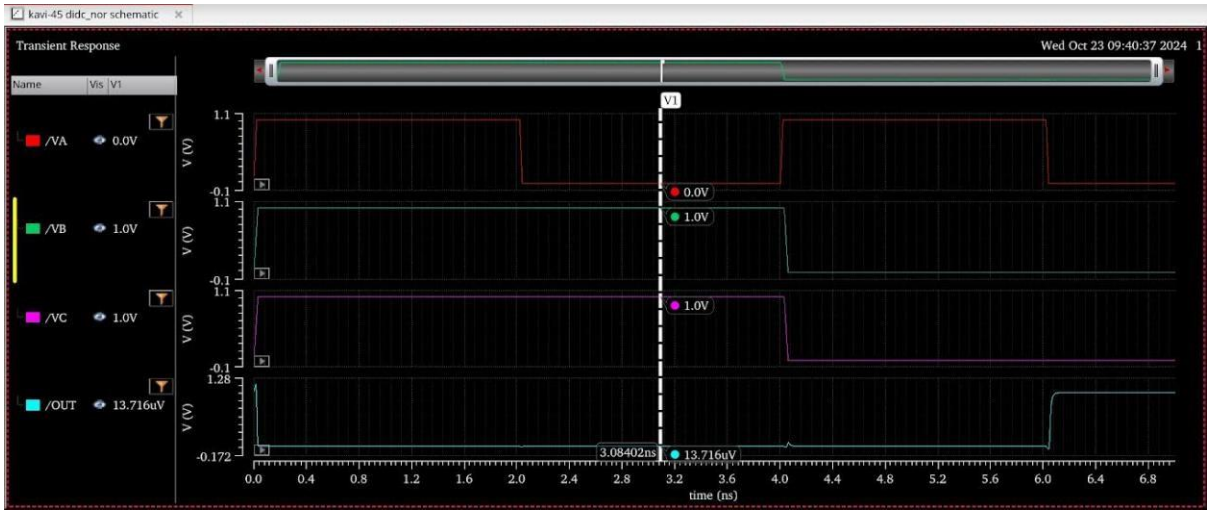


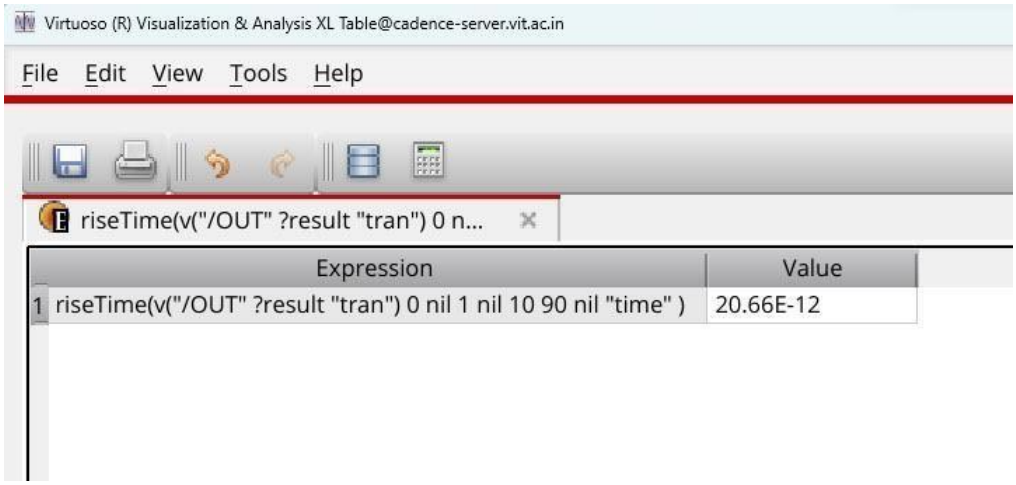
iii) 010




Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
20.36E-12

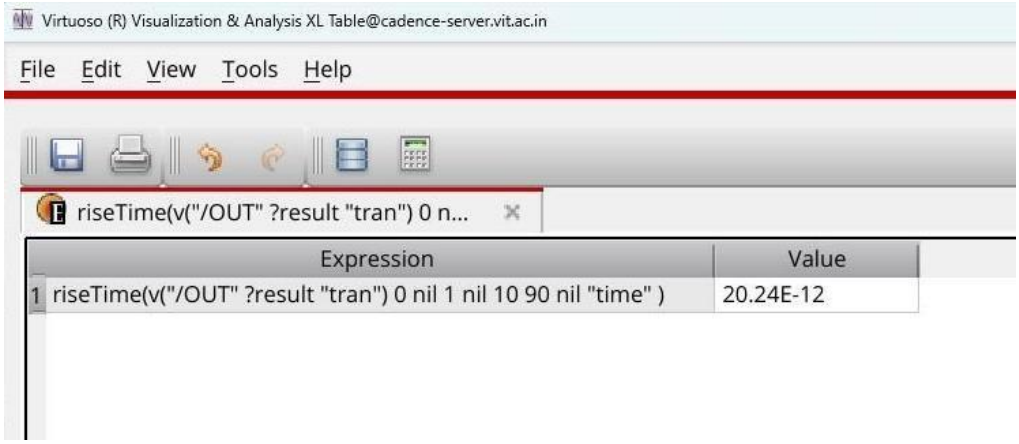
iv) 011


kavi-45 didc_nor schematic
Transient Response
Wed Oct 23 09:40:37 2024
Name
Vis
V1
/VA
0.0V
/VB
1.0V
/VC
1.0V
/OUT
13.716uV
V (V)
1.1
-0.1
1.28
-0.172
3.08402ns
13.716uV
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
4.0
4.4
4.8
5.2
5.6
6.0
6.4
6.8
time (ns)

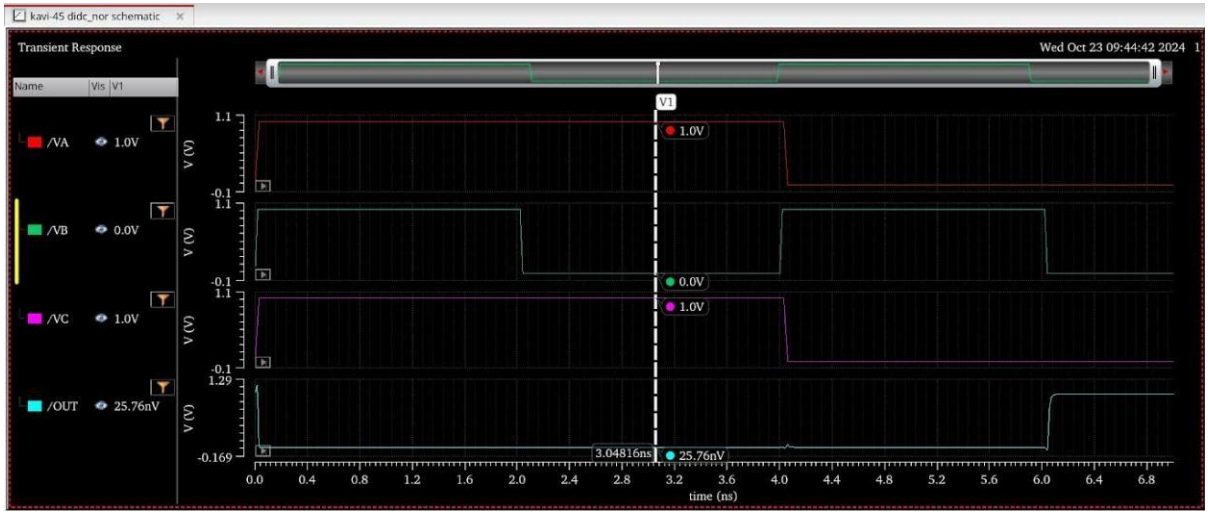

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
20.66E-12

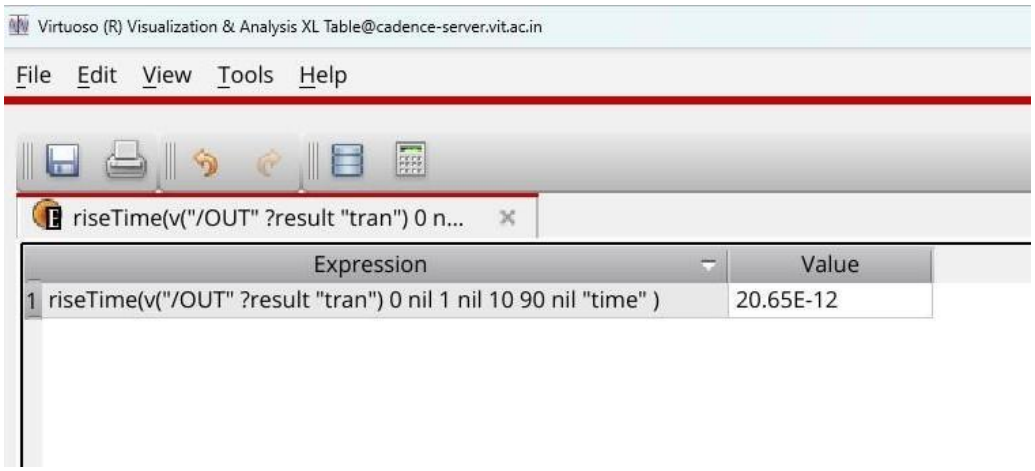
v) 100


kavi-45 didc_nor schematic
Transient Response
Wed Oct 23 09:43:04 2024
Name
Vis
V1
/VA
1.0V
/VB
0.0V
/VC
0.0V
/OUT
171.6uV
V (V)
1.1
-0.1
1.33
-0.31
3.08602ns
171.6uV
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
4.0
4.4
4.8
5.2
5.6
6.0
6.4
6.8
time (ns)

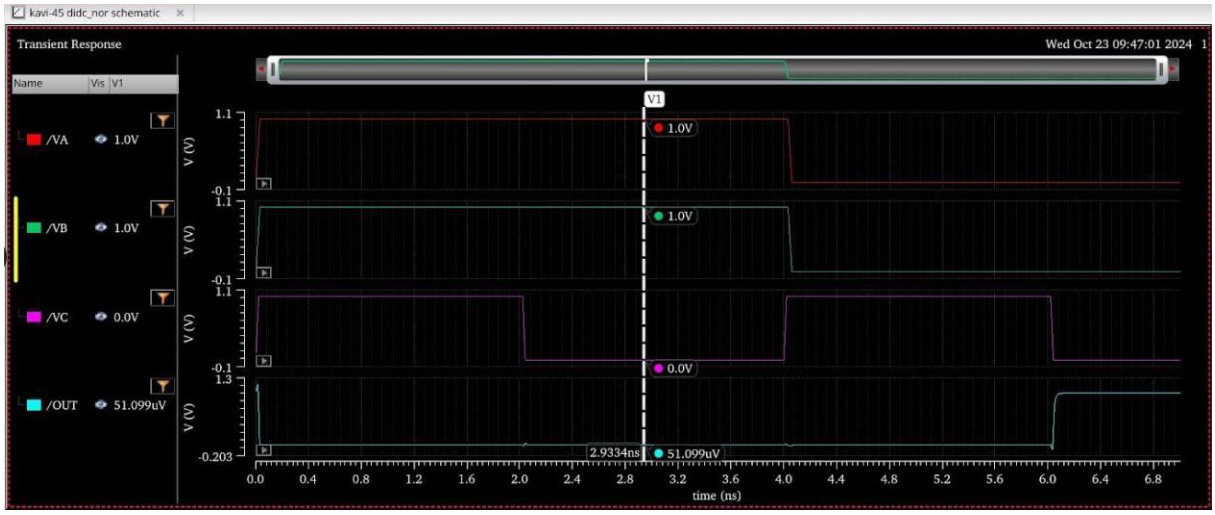

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File
Edit
View
Tools
Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1
riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
20.24E-12

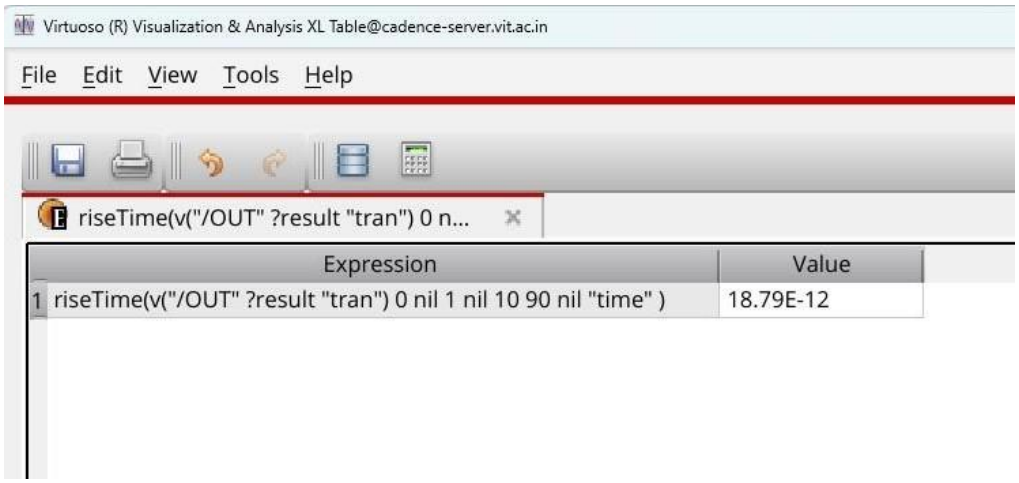
vi) 101


kavi-45 didc_nor schematic
Transient Response
Wed Oct 23 09:44:42 2024
Name
Vis
V1
/VA
1.0V
/VB
0.0V
/VC
1.0V
/OUT
25.76nV
V (V)
1.1
-0.1
1.29
-0.169
3.04816ns
25.76nV
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
4.0
4.4
4.8
5.2
5.6
6.0
6.4
6.8
time (ns)

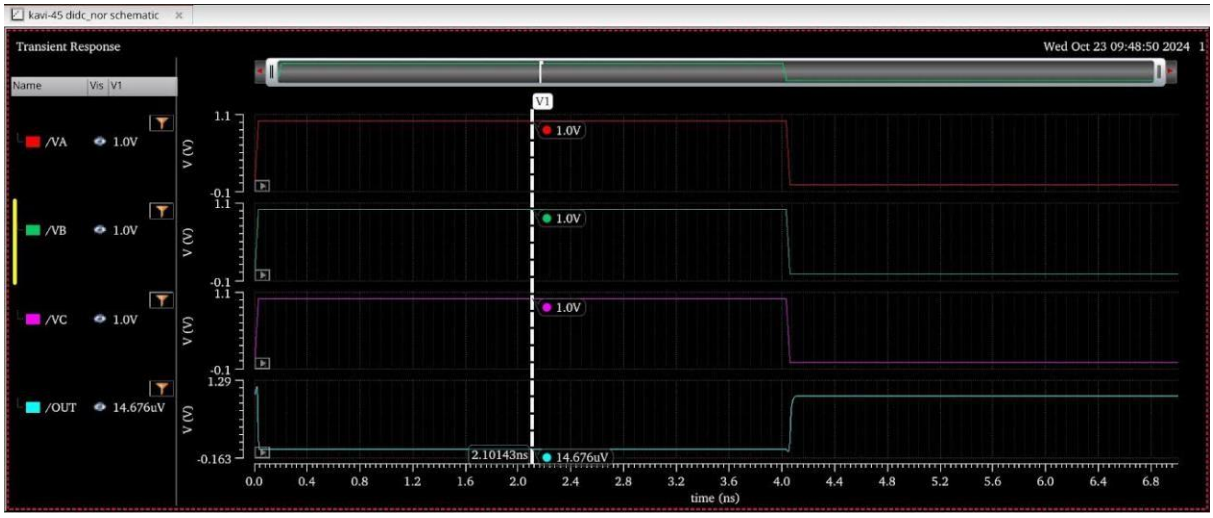

Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
20.65E-12

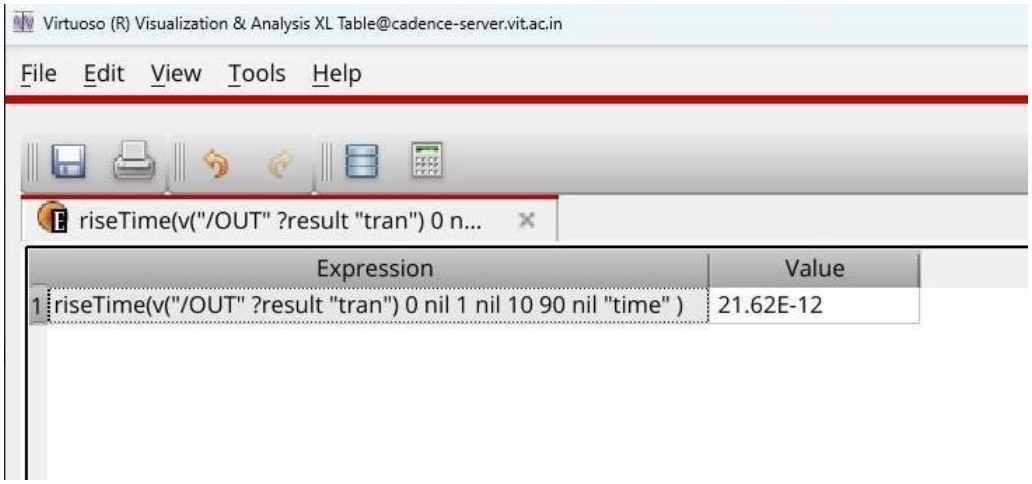
vii) 110


kavi-45 didc_nor schematic
Transient Response
Wed Oct 23 09:47:01 2024
Name
Vis
V1
/VA
1.0V
/VB
1.0V
/VC
0.0V
/OUT
51.099uV
V (V)
1.1
-0.1
1.3
-0.203
2.9334ns
51.099uV
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
4.0
4.4
4.8
5.2
5.6
6.0
6.4
6.8
time (ns)


Virtuoso (R) Visualization & Analysis XL Table@cadence-server.vit.ac.in
File Edit View Tools Help
riseTime(v("/OUT" ?result "tran") 0 n...
Expression
Value
1 riseTime(v("/OUT" ?result "tran") 0 nil 1 nil 10 90 nil "time")
18.79E-12

viii) 111





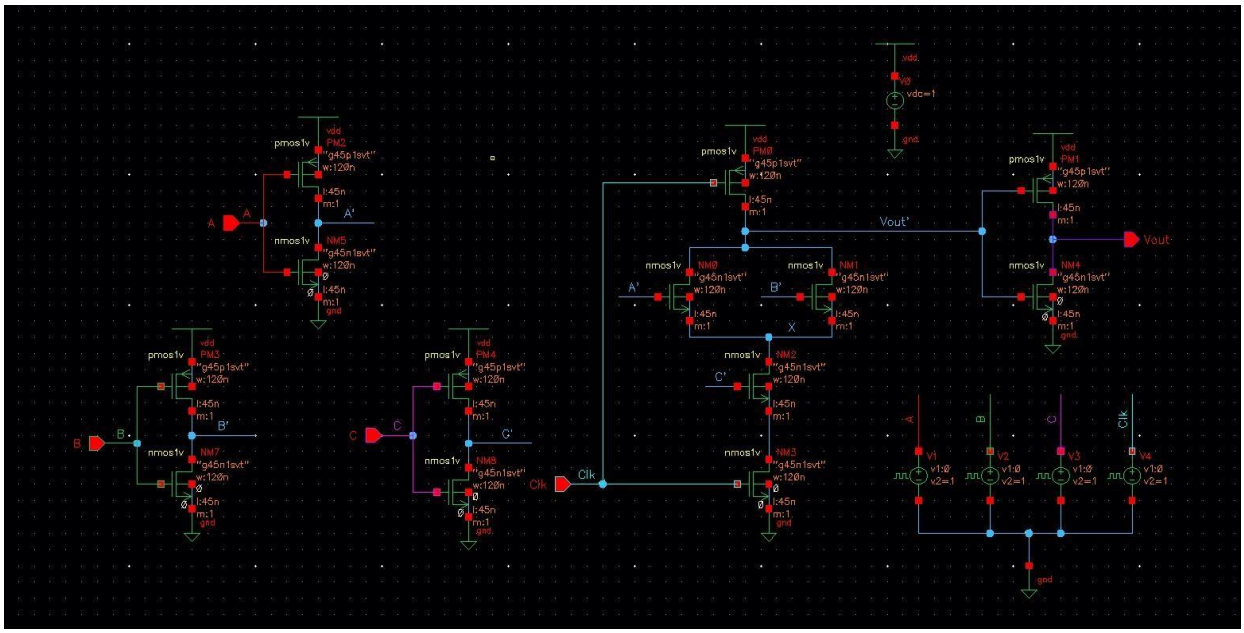
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3) Implement a Boolean function $\underline{Y = (AB+C)'}$ in domino logic and study the impact of the following through simulation.

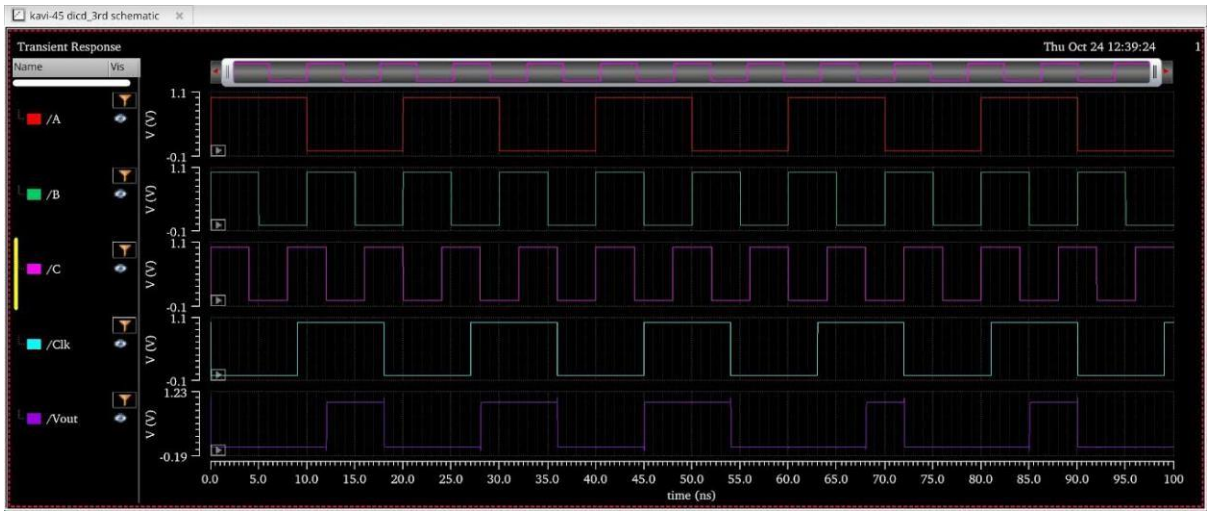
a) Sketch the Schematic and transient response
b) Effect of an input glitch on the output.
c) Impact of charge leakage
d) Charge sharing effect in dynamic logic.
e) Clock feed-through effect.

a) Sketch the Schematic and transient response

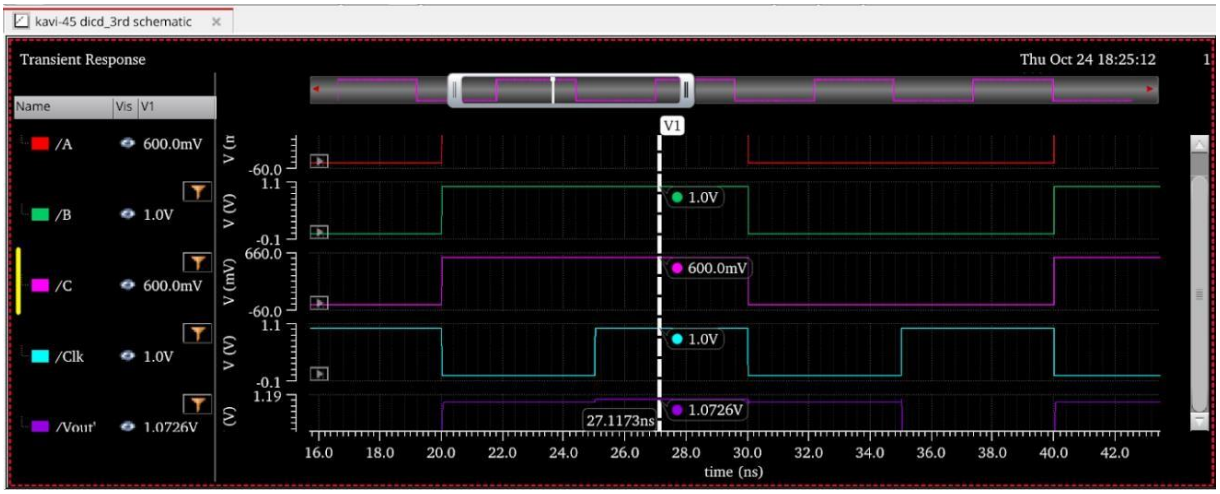
Schematic diagram:



Transient Response:



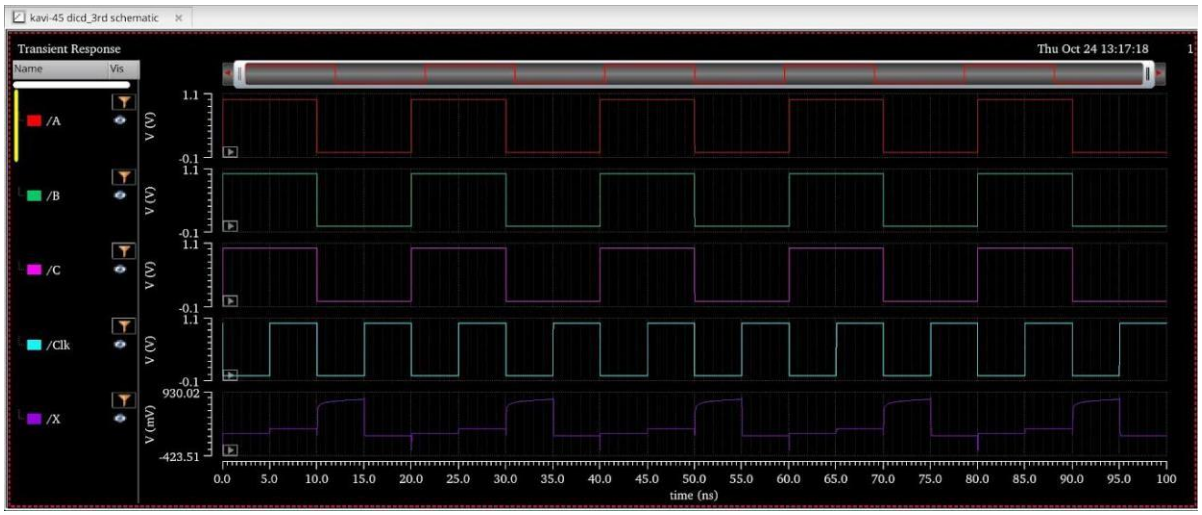
b) Effect of an input glitch on the output.



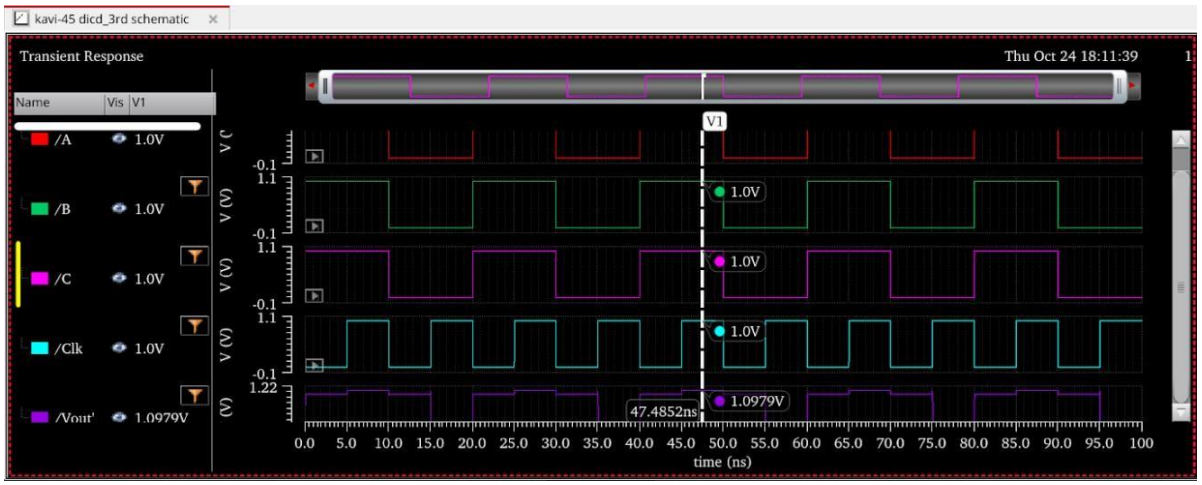
c) Impact of charge leakage



d) Charge sharing effect in dynamic logic



e) Clock feed-through effect.



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